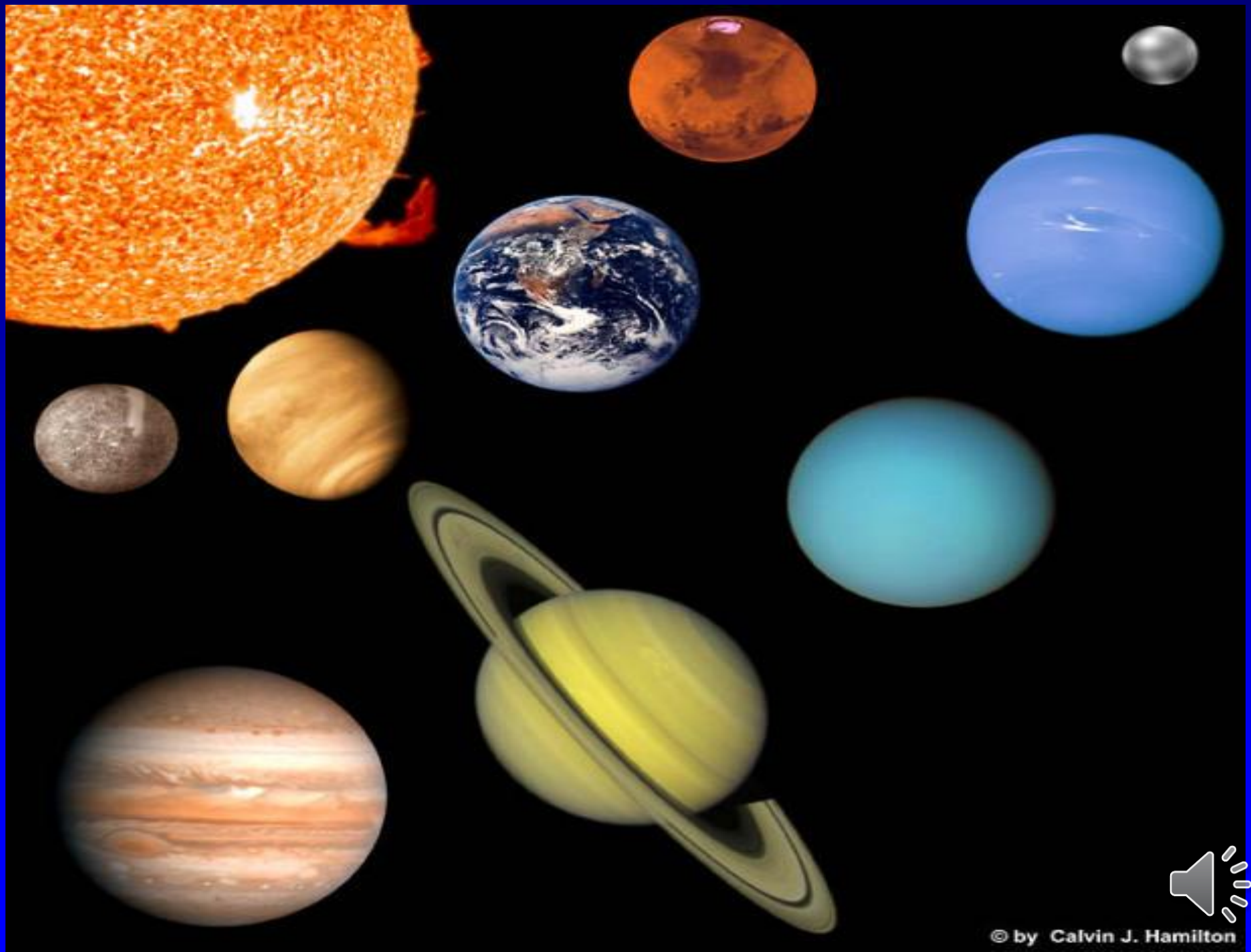
CHAPTER TWO

Solar System and the Earth

The Earth in Space

The Solar System that we live in consists of a medium-size star (the Sun) with eight planets orbiting it. The planets are of two different types;

- The inner planets (Terrestrial Plants)
  - Rocky planets near the Sun
  - Mercury, Venus, Earth & Mars
  - Arc, small, high density, rocky, poor in ices and H/He and no rings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The outer planets (**Jovian Planets)**

.Giant gaseous planets

.Jupiter, Saturn, Uranus, Neptune

. Large, low density, gaseous (H, He, CH_4), rotate rapidly.

. All have rings and have many moons

. Pluto, a small icy minor planet

- Differences between inner and outer planets:
  - Density
  - The rocky inner planets=densities of 3 g/cm^3 more
  - The gaseous outer Planets=densities of 1.6 g/cm^3 or less

ORIGIN OF THE EARTH AND THE SOLAR SYSTEM

The Nebular Theory

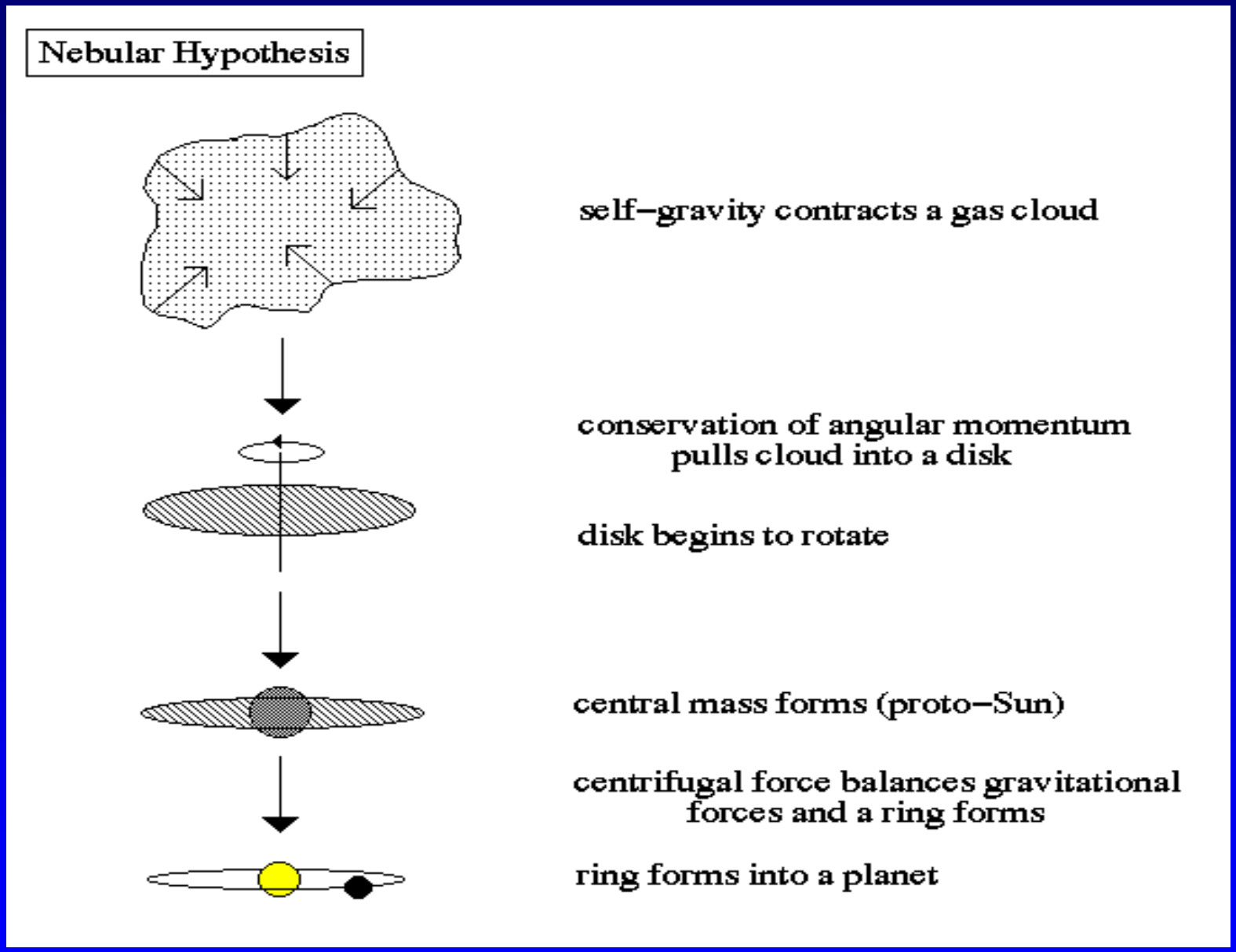


Fig. 1: Sketch of Nebular theory

How old of the Earth? Age of the Earth

4.5 to 4.6 billion years (4,500,000,000 to 4,600,000,000 years) as determined through radiometric dating (Uranium, Thorium), using an instrument called a mass spectrometer.

Where do we find the oldest rocks on Earth?

Canadian Shield. (NW Territories near Great Slave Lake, 3.96 by).

Before this, oldest rocks known were from Isukasia region of Greenland (3.8 by).

Spheres of the Earth

There are Four spheres of the earth: atmosphere, hydrosphere, biosphere and lithosphere.

Lithosphere: This crust is composed of igneous, sedimentary and metamorphic rocks.

Hydrosphere:

Total mass of water on or near the Earth's surface

-Covers 71% of Earth's surface

~98% in oceans

2% in glaciers, groundwater, lakes and streams (fresh water)

Biosphere:

It includes all life on the earth either plant life or animal life, in the sea or onland.

Atmosphere:

-78% nitrogen -21% oxygen

Minor amounts of carbon dioxide, argon and water vapor

- It is divided into several layers arranged upwards from the earth's surface:
- **Troposphere** (where weather takes place), **Stratosphere** (where the ozone layer that protects the surface of the Earth from UV rays is found),
- **Mesosphere, Thermosphere** and **Exosphere** (which grades off into outer space)

Earth Structure

How do we know what the Earth's Interior is like?

1.Drilling:

-Wells drilled into Earth are mostly in the upper 7 km of the crust. Deepest well- Soviet (Russian) well in northern Kola Peninsula 20 year effort to drill a 12 km hole and stopped in 1989. Costs are more than $100 million, bottom hole temperature is 190°C Current status?

2-Volcanic activity:

-Materials are brought up from below. Xenoliths = foreign rock (pieces of the mantle in lava) example: Only useful to depth of about 200 km.

3- Samples of the solar system (meteorites).

4- Study of seismic waves generated by earthquakes and nuclear explosions.

Ordinary Chondritic Meteorite

-composition of asteroids and the early solar system

-chondrules

-pyroxene chondrule in thinsection

The Earth is divided into three chemical layers: the **core,** the **mantle** and the **crust** (Fig.).

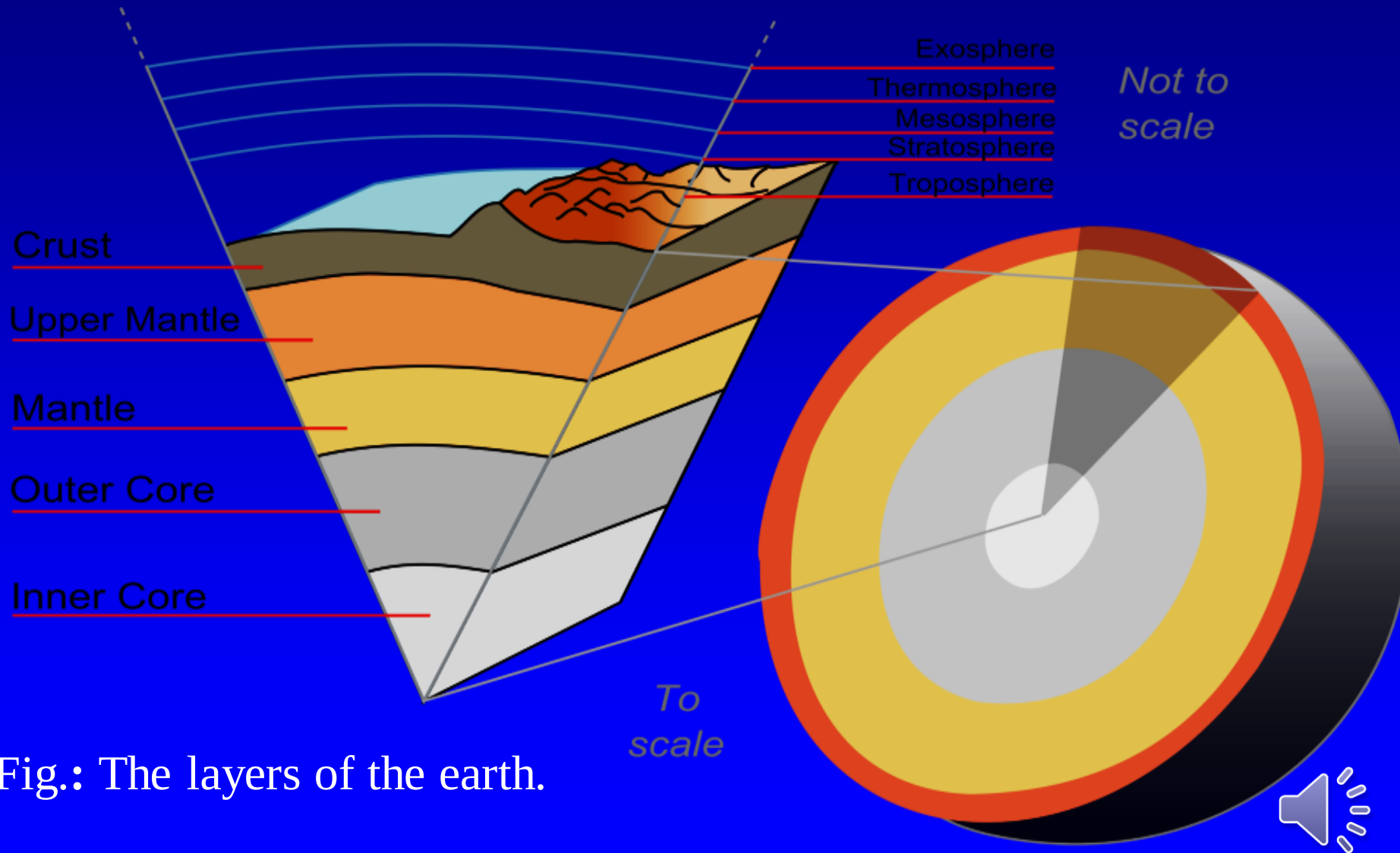
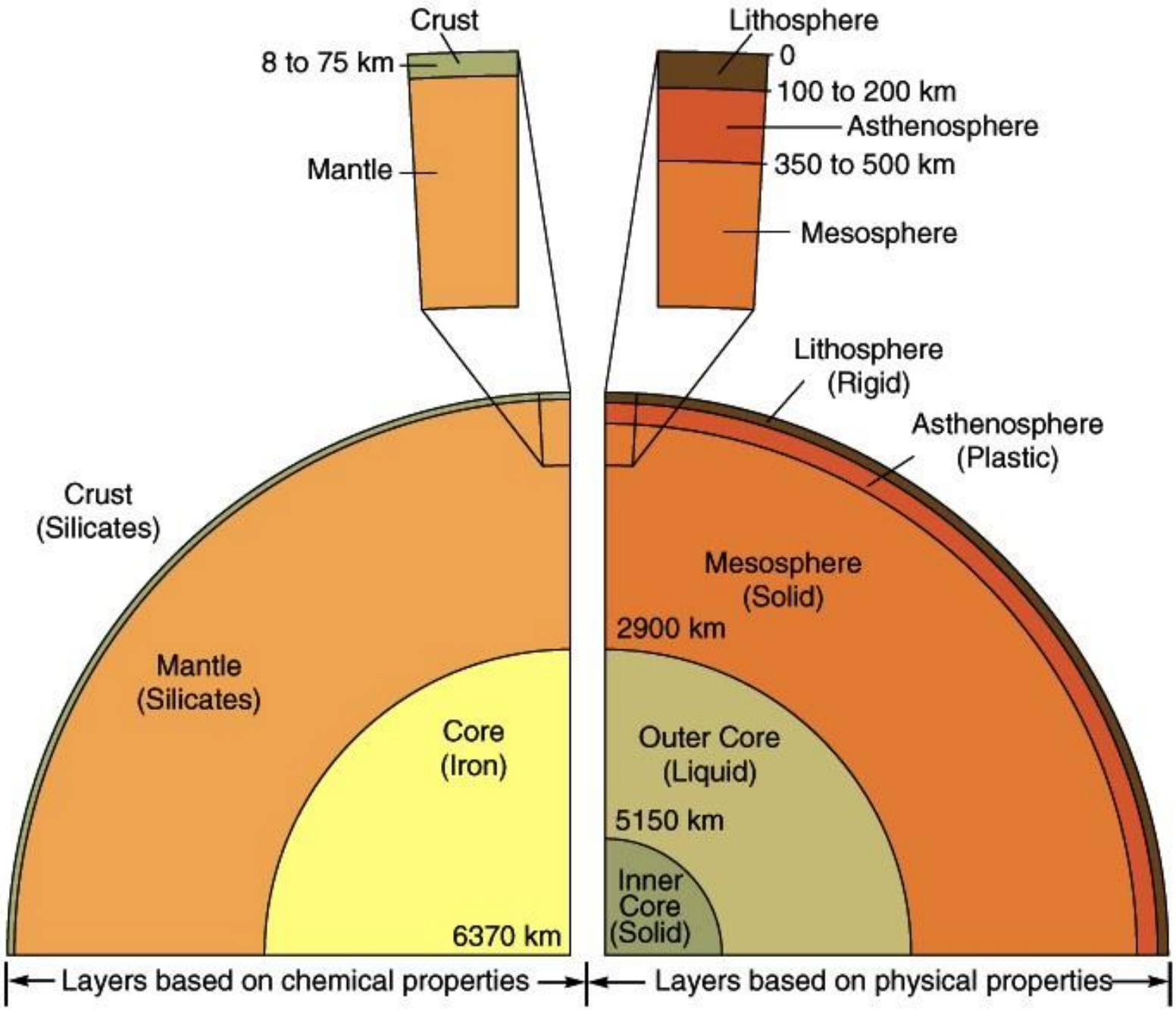
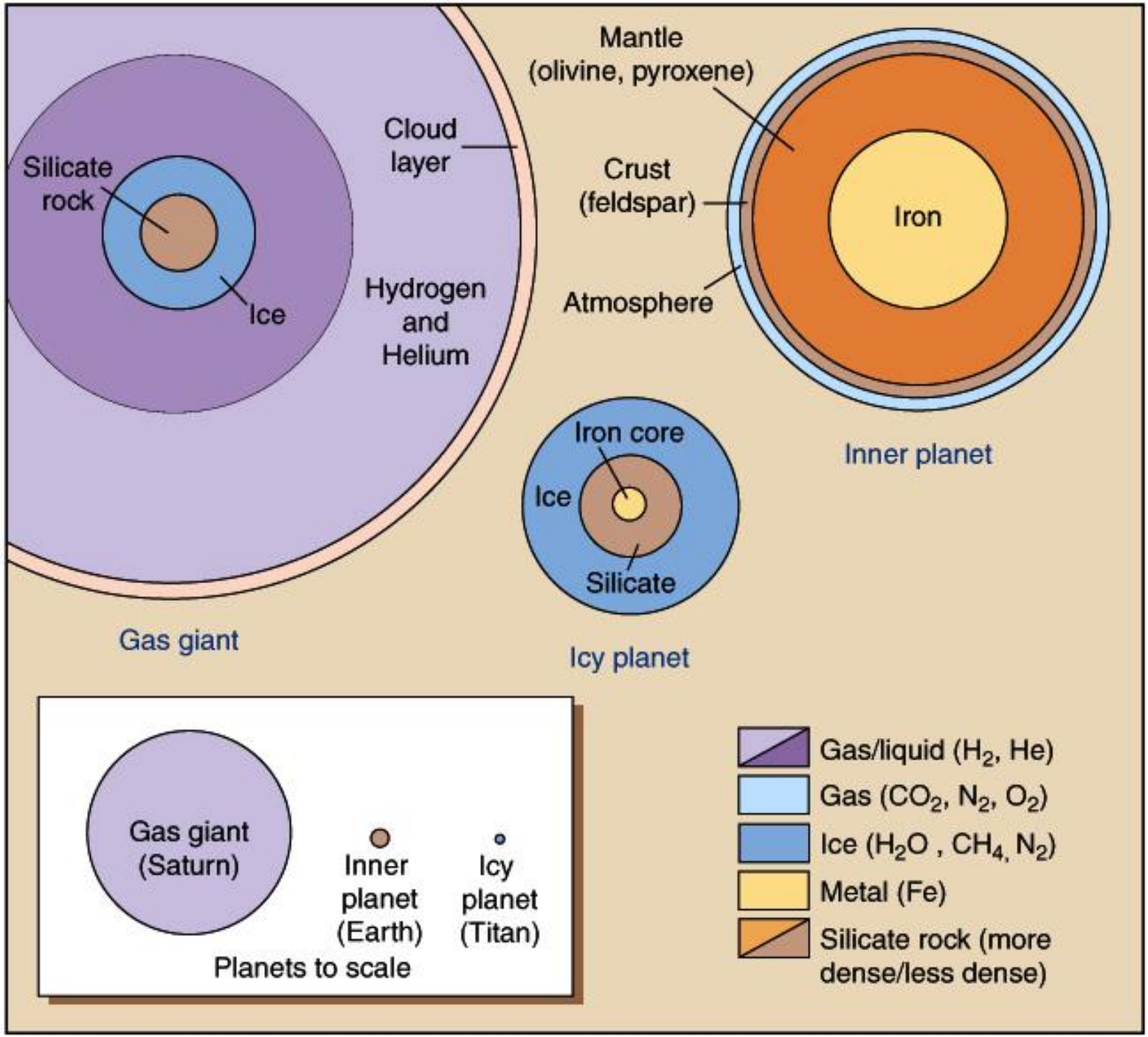


Fig.**:** The layers of the earth.

Earth's Internal Structure

- Solid Earth has a layered structure
  - Layers defined by composition and physical properties
  - **Compositional layers**
    - crust - mantle - core
  - **Physical layers**
    - lithosphere - asthenosphere - mesosphere - outer core - inner core


Crust
8 to 75 km
Mantle
Lithosphere
0
100 to 200 km
Asthenosphere
350 to 500 km
Mesosphere
Lithosphere
(Rigid)
Asthenosphere
(Plastic)
Crust
(Silicates)
Mantle
(Silicates)
Core
(Iron)
6370 km
Mesosphere
(Solid)
2900 km
Outer Core
(Liquid)
5150 km
Inner
Core
(Solid)
Layers based on chemical properties
Layers based on physical properties


Silicate rock
Ice
Cloud layer
Hydrogen and Helium
Gas giant
Mantle (olivine, pyroxene)
Crust (feldspar)
Atmosphere
Iron
Inner planet
Iron core
Ice
Silicate
Icy planet
Gas giant (Saturn)
Inner planet (Earth)
Icy planet (Titan)
Planets to scale
Gas/liquid (H_2, He)
Gas (CO_2, N_2, O_2)
Ice (H_2O , CH_4, N_2)
Metal (Fe)
Silicate rock (more dense/less dense)

Compositional Layers

- Crust
  - Outermost compositional layer
  - Definite change in composition at the base of the crust
  - 2 types:
    - Continental crust (Granite)
    - Oceanic crust (basalts)

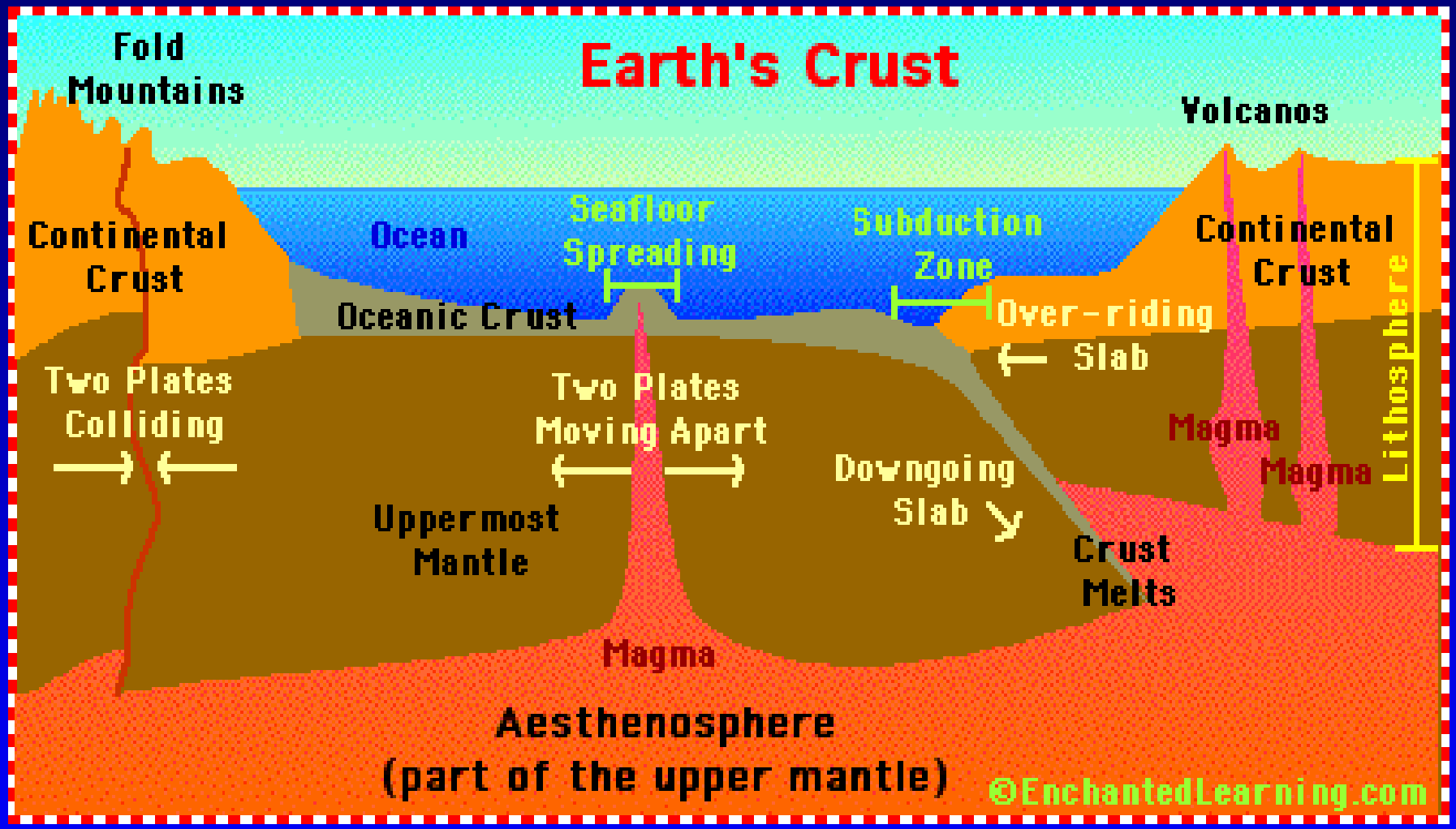
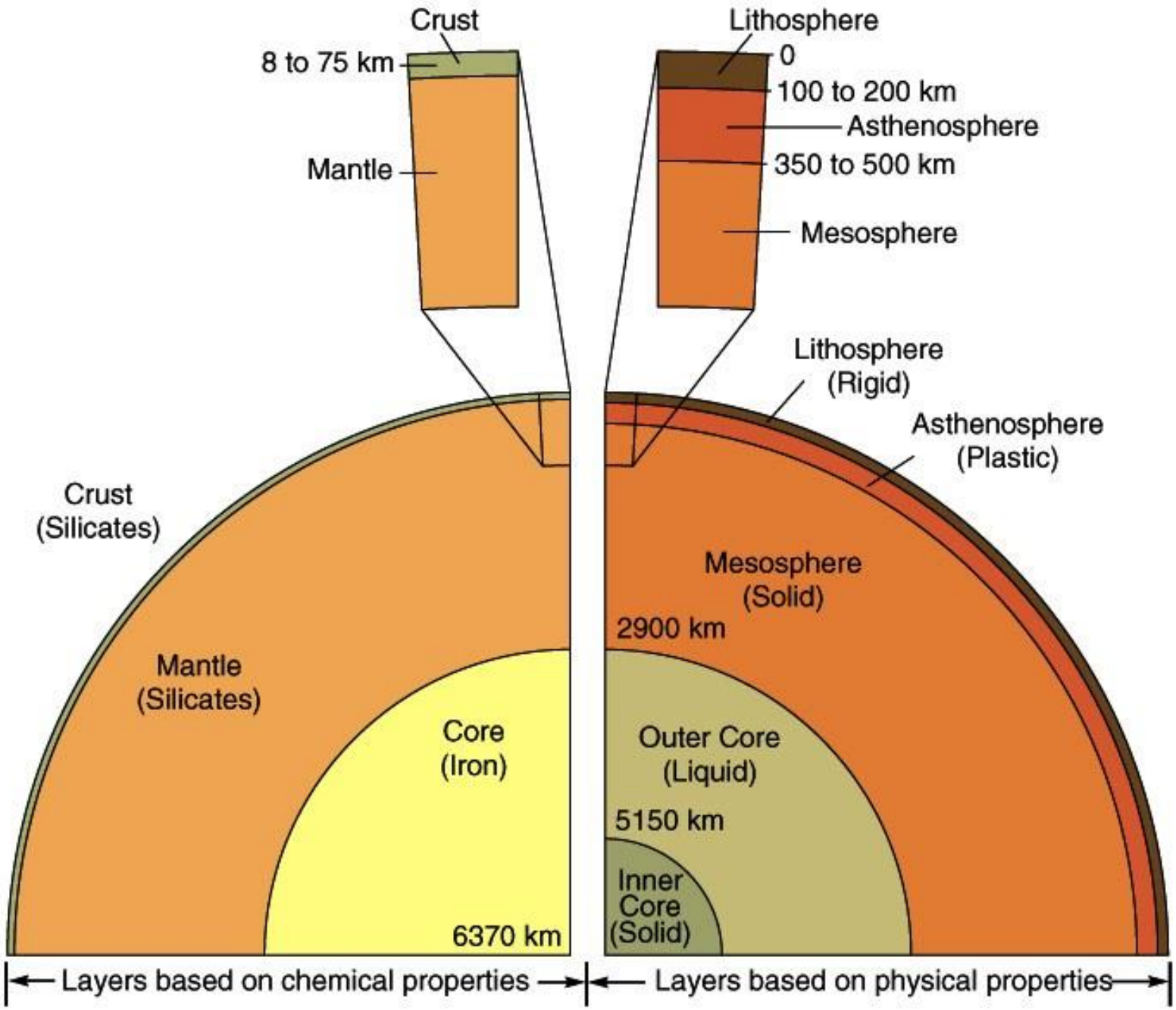


Fig. Lithosphere and asthenosphere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Layers based on chemical properties
Layers based on physical properties

- Crust
  - **Continental crust**
    - Thick - up to 75 km
    - Lower density - 2.7 g/cm^3
    - Strongly deformed
    - Much older - may be billions of years old

- Crust
  - **Oceanic crust**
    - Thinner - about 8 km
    - More dense - 3.0 g/cm^3
    - Comparatively undeformed
    - Much younger - < 200 million years old

- The asthenosphere is part of the mantle that flows. It might seem strange that a solid material can flow.

Physical Layers

- Lithosphere
  - Crust + upper portion of the mantle
  - Solid & rigid
  - Thickness ranges from 10 km beneath oceans to 300 km in continental areas

2- Mantle is the middle layer of the Earth, is made of minerals rich in the elements iron, magnesium, silicon, and oxygen.

Al over I000°C, the mantle is solid but can deform slowly in a **plastic manner.**

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The core is composed of iron and nickel and remains very hot, even after 4.5 billion years of cooling.

The core is divided into two layers: a **solid inner core** and a **liquid outer core.**
The inner Core is under such extreme pressure that it remains solid.

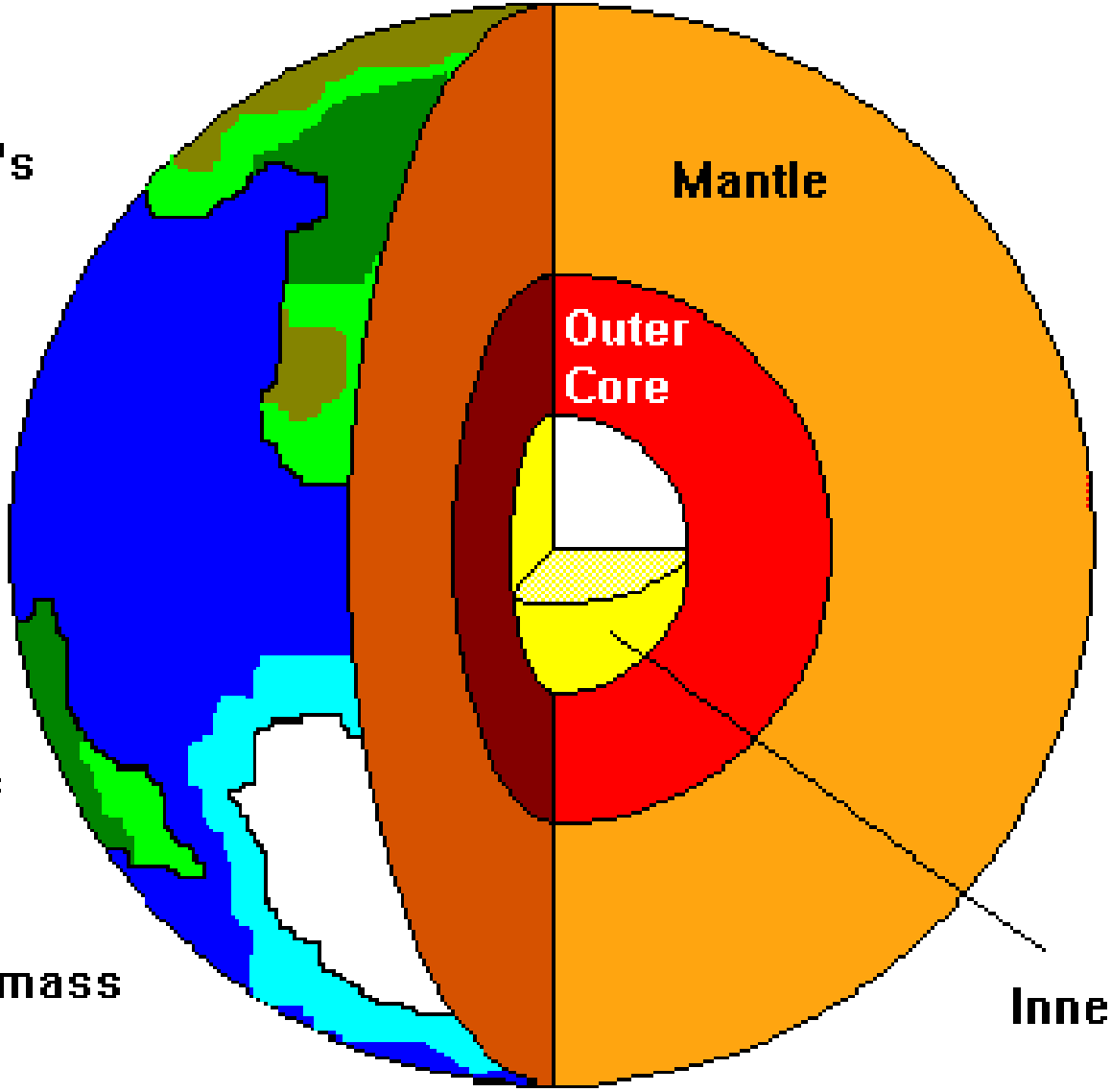
Atmosphere: 1/1,000,000 of Earth's mass

Oceans: 2/10,000 of Earth's mass

Crust: 1/250 of Earth's mass

Mantle: 2/3 of Earth's mass, 5/6 of volume

Core: 1/3 of Earth's mass 1/6 of volume



Exploring the Earth's Core

How was the Earth's core discovered?
Recordings of seismic waves from earthquakes gave the first clue.

* P and S wave travel times depend on properties of rock materials that they pass through. Wave velocity depends on density and elasticity of rock. Seismic waves travel faster in denser rock. Speed of seismic-waves increases with depth as pressure and density increase downward.